SRM Institute of Science and Technology

Course Code: 21CSC302J | Course Name: COMPUTER NETWORKS

Assignment 4

Unit-4: Medium Access Control

Instructions: Answer all questions. Draw neat diagrams wherever necessary. Show complete working for numerical questions.

Q1. Explain the ALOHA protocol. Differentiate between Pure ALOHA and Slotted ALOHA with their efficiency comparison.

Q2. Explain CSMA/CD (Carrier Sense Multiple Access with Collision Detection) with a flowchart. State where it is used.

Q3. Explain CSMA/CA (Collision Avoidance). Why is CSMA/CD not suitable for wireless networks?

Q4. Compare Ethernet and Token Ring technologies in terms of access method, topology and performance.

Q5. Explain Stop-and-Wait flow control with a neat timing diagram.

Q6. Explain Sliding Window flow control and state its advantage over Stop-and-Wait.

Q7. Explain Stop-and-Wait ARQ and Sliding Window ARQ as error control mechanisms, with diagrams showing retransmission on error.

Q8. Explain error detection techniques - Parity Check, Checksum and Cyclic Redundancy Check (CRC) - with one worked example each.

Q9. Explain Hamming code for single-bit error correction. Illustrate with a numerical example of encoding a 4-bit data word.

Q10. Compare HDLC (High-level Data Link Control) and PPP (Point-to-Point Protocol) as Data Link layer protocols.

--- End of Assignment ---